

11. Analysis B: effective resolution in H, V, Z

Lateral resolution is read from the depth ESF across each knife edge. Two estimates are made and cross-checked: one from the slanted-edge fit, one from the robot-stepped poses. Z resolution is the step size detected with 50% probability under the same false-alarm rule used in D.

11.1 H and V resolution

1. Reference levels. For each edge and frame, fit planes to the front and back surfaces, using only pixels farther from the edge than the 8 px analysis band (BOUNDARY_BAND_HALF_WIDTH_PX). These give Z_{front} and Z_{back} .
2. Normalized height. For each valid pixel, compute $h = (Z_{\text{back}} - Z) / (Z_{\text{back}} - Z_{\text{front}})$. Then $h = 1$ on the front surface and $h = 0$ on the back surface.
3. Signed distance. Compute each pixel center's signed distance s (px) to the true edge line, projected from the read-back pose carried through the registration and the as-built edge position, never from the commanded pose. s is positive on the front-surface side.
4. Robot-stepped ESF. Pool all frames and phase-jitter poses for that edge, Z , gap, and polarity. Bin by s in 0.25 px bins (BOUNDARY_BIN_WIDTH_PX), four per pixel. ESF(s) is the mean h of the valid pixels in each bin. Keep the per-bin count of no-reads for Section 14.
5. Slanted-edge ESF. On the nominal-pose mean frame, locate the half-height crossing in each row (or column) and fit a line to the crossings. Then bin by distance to that fitted line. Align the robot-stepped and slanted-edge ESFs at their half-height crossings; they should then agree to within one bin, and the residual difference is reported as an equivalent shift in px. If they do not agree, suspect the registration or a curved edge.
6. Metrics. The 10–90% rise distance is $s(h = 0.9) - s(h = 0.1)$, reported in px and in mm via $p(Z)$. The LSF is $d\text{ESF}/ds$, smoothed with a Hann window. The MTF is $|\text{FFT}(\text{LSF})|$ normalized to 1 at zero frequency. Report MTF50 in cycles/px. Rise distance is the primary figure, because a depth pipeline with interpolation is not a linear shift-invariant system, and an MTF assumes one.
7. Linearity test. Compare the ESFs at the 15 mm and 60 mm gaps (GAP_SMALL_MM, GAP_LARGE_MM). If they differ in h by more than 0.1 of the step height at any s (ESF_LINEARITY_TOLERANCE_H), mark the result step-height dependent and quote resolution together with G .

8. Edge offset. The offset between the fitted edge line and the true one, $s_{50} = s(h = 0.5)$, is the edge bias. A negative s_{50} means the measured edge lies on the back-surface side of the true edge, that is, the front surface is fattened. Section 14 reports s_{50} beside its near/far preference index.
9. Edge position transfer. For every phase-jitter pose of an edge, compute that pose's own edge crossing $s_{50}(\text{pose})$ from its ESF and plot it against the pose's read-back lateral offset across the edge, carried through the registration, in px. Because s is measured from each pose's own true edge, an ideal sensor gives the same s_{50} at every offset; regress $s_{50}(\text{pose})$ on the offset in a joint fit with the periodic terms below: the lateral gain is 1 plus the slope, expected to be 1, and the intercept is the edge bias of Step 8. Fit the residual with a periodic term of 1 px period (pixel locking) and one at the projector dot pitch taken from the correlation length of Section 10, Step 7, and report both amplitudes with the gain for each Z and orientation. Where the optional lateral sweep of Section 6.1, Step 6 was run, add the approach hysteresis, the mean difference of s_{50} between the poses approached from the negative and from the positive side, and refit the periodic terms at the sweep's 0.1 px spacing. This uses the B captures as they are and adds no capture.
10. Report for each Z, orientation (H or V), polarity, and gap. Confidence intervals come from bootstrapping over poses with 2000 resamples (BOOTSTRAP_RESAMPLES).

11.2 Z resolution

11. Sensed step. For each consecutive $A \rightarrow B$ visit pair, compute $\Delta = \text{mean}(Z \text{ at } B) - \text{mean}(Z \text{ at } A)$. Do this for patches of 1 px, 5×5 px, and 20×20 px (ZSTEP_PATCH_SIZES_PX), tiled over the ROI. The true step δ is the difference of the read-back target pose along the optical axis between the two visits; the registration's translation cancels in the difference, so only the robot's relative motion accuracy enters. A rung smaller than twice the robot repeatability of 0.05 mm (ROBOT_REPEATABILITY_MM), which equals the 0.1 mm floor of the ladder of Section 6.2 at the indicative values, is flagged as unreliable truth and left out of the gain regression of Step 12; at the indicative values none is flagged, and the rule matters for a less repeatable robot. Also report the scatter of the read-back Z over the $A \rightarrow A$ pairs as the robot's own reported repeatability.
12. Gain and linearity. Regress the mean Δ on δ across the ladder. Report the slope (gain), the intercept, and the largest deviation from the line.
13. Null distribution and threshold. Differences between consecutive A visits ($A \rightarrow A$) form the no-step distribution for each patch size. Set τ at its 99% quantile, one minus the 1% false-alarm target (DETECTION_FALSE_ALARM_TARGET).
14. Detection curve. A step counts as detected if $\Delta > \tau$ in the commanded direction. Fit the detection fraction against $\log \delta$ with the psychometric model of Section 13. Report $\delta_{50}(Z, \text{patch})$ with its confidence interval. For uncorrelated noise, δ_{50} shrinks as the square root of patch area.

Shrinking more slowly than that means the noise is correlated over that scale, consistent with the correlation length from Section 10.

15. Ramp. For each ramp pose, subtract the fixed-pattern map of A at the same station, then average the depth along each image row within the plate's region of interest. The true depth of each row comes from the read-back pose of the tilted plate carried through the registration. Plot the row average against the true depth. Where the curve is stepped, the plateau widths give the measured quantum $\delta Z_q(Z)$; compare it with $q \cdot Z^2/k$ from Section 10. Plot single pixels the same way. The row average shows plateaus only when the disparity noise is well below the quantum; at the indicative noise of about 0.6 quantum the frames dither the quantizer, the averaged curve is smooth at the near stations, and plateaus are resolved only at the far ones. A curve counts as stepped when fewer than half of its changes over a half-quantum window (`RAMP_MAX_INTERMEDIATE_FRACTION`) fall between a quarter and three quarters of a quantum (`RAMP_INTERMEDIATE_BAND`), and smooth otherwise; a smooth row average over stepped single pixels is reported as a finding, dithering by the sensor's interpolation or noise, not as a failure. For a smooth curve the quantum comes from the spacing of the populated depth levels of the pooled ramp readings, the method of Section 10, Step 8, and the output says which method gave it. Where the optional staircase was run, plot its sensed Z against the read-back Z and compare the two quanta.
16. Outputs. `B_resolution_summary.csv` with one row per edge, Z, orientation, polarity, and gap, including the edge position transfer terms; `Z_resolution_summary.csv` and `Z_resolution_rungs.csv` for the ladder; `Z_ramp_rows.csv` for the ramp; `B_lsf.csv` with the H and V line spread functions per edge and Z, for Section 12. Figures (PNG and SVG): ESF curves; LSF and MTF; rise distance and `s_50` against Z; the edge position transfer plots; Z-step detection curves and sensed against true step; the ramp rows and the quantum against Z. The rise distances in H and V and the mean `s_50` go in `forward_model_parameters.json` for the Tier-A simulator.